

**UART Based Communication Between Two MSP430
Microcontrollers
BY: Muhammad Haseeb**

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1 Objective

The objective of this project is to design and implement a UART (Universal Asynchronous Receiver Transmitter) based communication system using two MSP430F249 microcontrollers.

One MSP430 acts as a transmitter that continuously sends the predefined string

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while the second MSP430 acts as a receiver that receives the transmitted characters and displays the complete message on a 16×2 LCD module.

The project demonstrates serial communication, UART configuration, LCD interfacing, interrupt-based reception, and embedded system communication.

2 System Overview

The proposed system consists of two MSP430F249 microcontrollers connected through UART communication.

- MSP430-1 operates as the UART transmitter.
- MSP430-2 operates as the UART receiver.
- UART communication is established by connecting:
 - TX (P3.4) of transmitter to RX (P3.5) of receiver – RX (P3.5) of transmitter to TX (P3.4) of receiver
 - Common Ground
- The receiver stores incoming characters into a buffer.
- When a newline character is received, the complete string is displayed on the LCD.

Data Flow



3 Schematic Design

Figure 1 shows the complete Proteus simulation of the project.

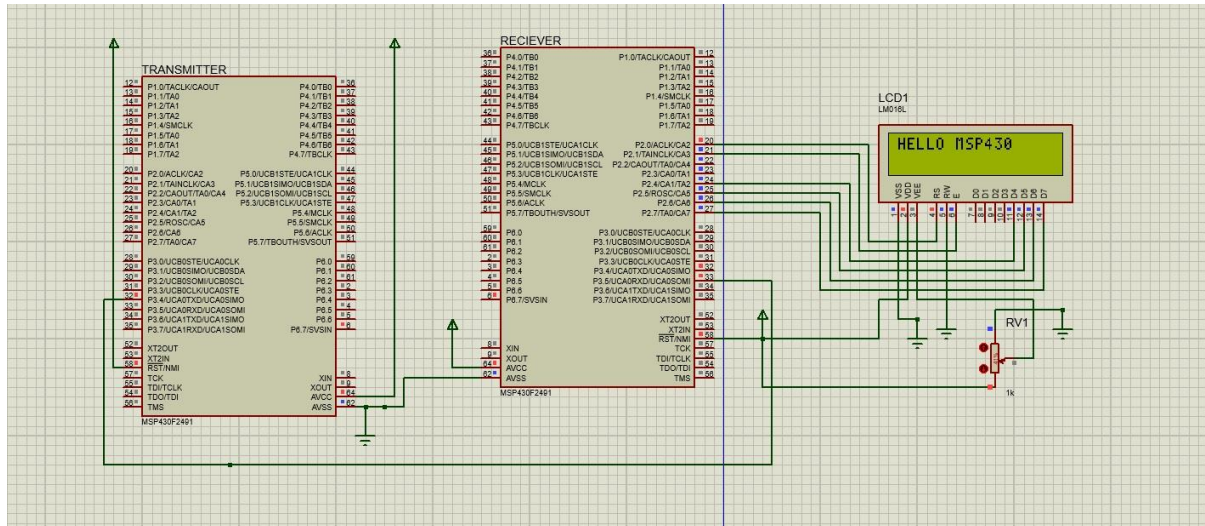


Figure 1: Complete UART Communication System in Proteus

The schematic contains:

- Two MSP430F249 microcontrollers.
- UART cross connection between both controllers.
- Common ground.
- 16x2 LCD connected to the receiver.
- LCD contrast control using a potentiometer.
- Power supply connections.

The transmitter continuously sends the predefined string through UART. The receiver captures each character using the UART receive interrupt and updates the LCD after receiving the newline character.

4 UART Communication Design

UART (Universal Asynchronous Receiver Transmitter) is an asynchronous serial communication protocol that transfers data without a clock signal.

The following configuration is used:

Clock Source	SMCLK (1 MHz)
Baud Rate	9600 bps

Data Bits	8
Stop Bits	1
Parity	None
Mode	Asynchronous

The UART registers are configured as follows:

- **UCA0CTL1** selects SMCLK as UART clock source.
- **UCA0BR0 = 104**
Used for generating a baud rate of 9600 bps from a 1 MHz clock.
- **UCA0BR1 = 0**
High byte of baud rate divider.
- **UCA0MCTL = UCBSR0**
Provides baud-rate modulation for improved timing accuracy.
- UART pins are enabled by

$P3SEL \mid = \text{BIT4} + \text{BIT5};$

where

- P3.4 = TXD
- P3.5 = RXD

5 LCD Interfacing Design

The receiver interfaces a standard 16×2 LCD in 4-bit mode.

The LCD uses only four data lines, reducing the number of GPIO pins required.

The LCD initialization sequence includes

- Enter 4-bit mode
- Display ON
- Cursor increment

- Clear display

Whenever a complete string is received, the receiver

1. Clears the LCD.
2. Moves the cursor to the first position.
3. Displays each received character.

6 Software Design

The software is divided into two programs.

6.1 Transmitter

The transmitter performs the following steps:

1. Initialize system clock.
2. Initialize UART.
3. Continuously transmit

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one character at a time.

4. Send newline character.
5. Wait before retransmitting.

6.2 Receiver

The receiver performs:

1. Initialize LCD.
2. Initialize UART.
3. Enable UART Receive Interrupt.
4. Store received characters into a buffer.

5. Detect newline.
6. Display received string on LCD.

7 Code Structure

The project is divided into modular functions.

UART init()

Initializes UART peripheral using SMCLK at 9600 baud.

UART sendChar()

Transmits one character after checking that the transmit buffer is ready.

LCD init()

Initializes the LCD in 4-bit mode.

LCD cmd()

Sends command bytes to the LCD.

LCD data()

Displays one character on the LCD.

RX ISR

The UART receive interrupt automatically stores incoming characters into the receive buffer.

8 Testing and Results

The system was simulated successfully in Proteus.

Observed results include:

- UART communication established successfully.
- Characters received correctly.

- No missing characters.
- LCD displayed

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- Continuous transmission worked without errors.

9 Challenges and Solutions

Challenge 1

Initially, no data was transmitted.

Solution

The UART module remained in software reset mode. The instruction

`UCA0CTL1 &= ~UCSWRST;` released the UART from reset,

allowing communication.

Challenge 2

The LCD displayed incomplete or corrupted characters.

Solution

Proper delays were added after every LCD command and data write, and the LCD initialization sequence was corrected for reliable operation.

10 Advantages

- Simple hardware.
- Reliable asynchronous communication.
- Low power consumption.
- Interrupt-based reception.
- Easy expansion for sensors and wireless modules.

11 Applications

- Embedded monitoring systems
- Industrial automation
- IoT devices
- Sensor data transmission
- Robotics
- Serial debugging

12 Conclusion

This project successfully demonstrates UART communication between two MSP430F249 microcontrollers.

The transmitter continuously sends the message “HELLO MSP430” through UART at 9600 baud, while the receiver collects the transmitted characters using interrupt-driven reception and displays the message on a 16×2 LCD.

The project provides practical understanding of UART communication, interrupt handling, LCD interfacing, and embedded system design. It also demonstrates reliable serial communication between multiple microcontrollers, which forms the basis of many industrial and IoT applications.